

NAME _____

ChE 210

Quiz #1

4 February 2020

Experimental data for a liquid/vapor binary mixture of pentane (P) and heptane (H) is given in the table. The quantity z_{liq} gives the mole fraction of pentane in the liquid phase while z_{vap} gives the mole fraction of pentane in the vapor phase. The data can be linearized using the relation $Y = A X$ where A is the separation factor, $X = z_{liq}/(1 - z_{liq})$, and $Y = z_{vap}/(1 - z_{vap})$.

For a general straight line with the intercept equal to zero $y = a x$, the “best fit” slope for perpendicular offsets is given by $a = -\bar{B} \pm \bar{B} \sqrt{1 + 1/\bar{B}^2}$ where $\bar{B} = \frac{S_{xx} - S_{yy}}{2S_{xy}}$ and

the averages are defined as $S_{xx} = \frac{1}{n} \sum_{i=1}^n x_i^2$, $S_{yy} = \frac{1}{n} \sum_{i=1}^n y_i^2$, and $S_{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i$, respectively. If $S_{xx} - S_{yy} > 0$, use the plus sign in front of the square root in the expression for the slope. If $S_{xx} - S_{yy} < 0$, use the minus sign.

(a) Give the units on the mole fractions z_{liq} and z_{vap} .

(b) Give the units on the mole fraction ratios X and Y .

(c) Calculate the “best fit” value of the separation factor using perpendicular offsets as described above. Show all of your work.

Mole Fraction z_{liq}	Mole Fraction z_{vap}
0.867	0.984
0.594	0.925
0.398	0.836
0.254	0.701
0.145	0.521
0.059	0.271